

# PUBLICATION POLICY FOR THE JADES COLLABORATION

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|---------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| 071     | Nov 2023   | Minor revision about how to start CWR                                                                                                                                                 |

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## Introduction

This document describes the policies and guidelines governing the publication of scientific and technical results from the JWST Advanced Extragalactic Survey (JADES) collaboration. The policy is to maximize the scientific output of JADES, to promote the scientific excellence and impact of the resulting publications, as well as their technical accuracy and integrity.

The policy aims to provide opportunities to all JADES team members including junior members to contribute to the various scientific investigations in a coordinated and timely manner. At the initial stages of the JADES collaboration, lead author(s) were to come from the 'scientifically primary' instrument team. However, this new edition of the publication policy outlines how the collaboration will move forward with a more flexible policy allowing greater collaboration between NIRCam and NIRSpect teams. Reserved abstracts from each instrument team from the 'first wave' of science papers are still active, and as such the information on the agreed straw-man distribution of leadership per scientific question is presented in Appendix A.

The policy also aims to make sure that fair and proper credit is given to individuals for their work in the team.

The policy applies to all members of the JADES collaboration, as established in the document describing the organization of the JADES collaboration.

## Scope

The present policy applies broadly to: papers submitted to archives and refereed journals, books or book sections about JADES and its results, graphics or tabular materials or discussions of results, in electronic or hard-copy form, that are based on JADES preparatory work, analysis of its data or on JADES tools / infrastructure and are accessible to non-JADES team members.

This policy does not cover education and public outreach (EPO) communications and publications; for these, SC approval is required for disclosure of non-public data.

As JADES data becomes public, we stress the following aspects:

- The JADES Publication Policy applies whenever the work begins before the data release.
- The JADES Publication Policy applies when any of the data used remains private (such as intermediate data products from collaboration pipelines).
- The JADES Publication Policy applies when the work is using internal resources, such as non-public FITSmaps or other data exploration tools.
- The JADES Publication Policy does not necessarily apply to cases where the work is based purely on public JADES data and starts after the data release; however, entanglements can happen when the work is benefitting from discussions within JADES communication channels, from exposure to unpublished results, or from other JADES products that remain proprietary. If JADES members plan to work on public JADES data in a manner separate from the collaboration, it is important to disclose this to the group. Please talk to the Steering Committee regarding such sensitivities.

## The Publication Coordinators

Responsibility for oversight of the JADES publications will rest with the JADES steering committee. The JADES PIs will select one or two JADES collaboration members as the JADES Publication Coordinator (PC) who will act as the primary point of contact for the collaboration for any publication-related topic. Each Publication Coordinator is appointed for a 1-year term, renewable. The Publication Coordinators are a non-voting, ex-officio member of the JADES steering committee.

The Publication Coordinator(s) will:

- Be responsible for the implementation of the publication policy and in particular of making sure the procedures are implemented and followed.
- As described in more details in the procedures, verify consistency with the collaboration agreements, check for overlap between publications, and if necessary work with concerned parties to mitigate them.
- Maintain up-to-date lists of the published and in-preparation articles of the JADES collaboration on the JADES Confluence/Jira site.
- When relevant, propose revisions of the publication policy in consultation with the JADES collaboration. Final approval of the changes rests with the steering committee.

## Papers and Projects

JADES maintains a set of a few dozen “projects” to coordinate its work. Projects are long-standing efforts to tackle key science themes and infrastructure. Projects are led by one or two convenors, appointed by the SC. The SC can create or disband projects as needed, and projects may evolve as the overall JADES program matures with increasing collaboration between the original instrument teams.

“Papers” then refer to individual journal papers, or equivalent products. Papers will usually be associated to one or more projects; if no project matches, then the SC will function in the Project Lead role. It may happen that an analysis effort will result in multiple papers; we ask that these be declared separately so they can be tracked. It can also happen that the realization that a single analysis would be better presented as multiple papers comes late in the process; this is fine and would result in a declaration of the additional paper(s).

## Paper categories

We define a set of paper categories:

- “key paper”: these papers, identified as flagship JADES papers by the PIs will have an enlarged list of authors. Only a limited number of papers will belong to this category.
- “standard paper”: default category, for papers with standard authorship rules and timeline.
- “fast-track paper”: papers with a reduced timeline. Moving a paper to this category can be requested by the lead author, typically when the paper is nearly ready for collaboration-wide review. A clear and compelling justification for the urgency of release must be given. The steering committee must approve such designation.

## Declaring a planned paper

Papers are declared by their lead author, in consultation with the relevant Project Leads and Publication Coordinator(s), with approval coming from the SC. Papers should be declared as early as possible in the life cycle of the analysis, as these declarations are the main way that the collaboration seeks to minimize conflict between work and to identify opportunities that may be under-served. Declarations also help the lead author(s) clarify the scope of their paper, get feedback about strategies to speed the work to completion, and get in the queue to give updates to the broader collaboration.

It is understood that for cases where the lead author is a student, their advisor may end up conducting some of the lead author responsibilities listed below.

### Procedure

1. The lead author drafts a proposal, including title, abstract, lead author, list of other core contributors, lists of external collaborators, associated Projects, intended data set, declaration of availability of effort, and goal completion date. The title and abstract should be descriptive of the planned work; it is not expected that these will match to what the final manuscript uses. Dependencies on external collaborators must be documented. Further information on the template is in Appendix B.
2. The lead author circulates the proposal to the relevant Project Leads, who will advise regarding possible overlaps with previously approved papers and help to negotiate any sensitivities. The Project Leads will also advise regarding sharpening the focus and clarity of the abstract; it is desirable that individual papers be tightly focused, to make them easier to finish and to leave more room for other people's work. It is important that abstracts plan a scope that can reasonably be presented in a single paper. If necessary, the lead author and/or Project Leads should engage the Publication Coordinators for advice if difficulties are arising.
3. After this consultation, the lead author circulates the proposal to the collaboration, using the general email list and the #general channel on Slack. The Publication Coordinator will then schedule a discussion in the next pan-JADES telecon, where the lead author will present the proposal to the collaboration. This may surface additional suggestions or expressions of concern about interfaces to other work. JADES strongly encourages members to express concerns when they feel that a conflict between two efforts is looming; it is far better to surface these issues early. After the telecon, the lead author will revise the proposal, consulting with the Project Leads and Publication Coordinators as needed.
4. The lead author sends the final proposal to the Steering Committee for approval. The SC will request comment from the Project Leads regarding how the proposal fits into the general progress of the associated project(s) and how any concerns/conflicts have been addressed. Following this, the SC will vote regarding approval.
5. The Publication Coordinator will communicate the decision to the lead author and add the paper to the JADES list of approved papers.

JADES members can be a lead on no more than 3 papers at any one time, and any paper beyond the first carries the expectation that the lead author clearly document how the needed effort will be marshaled or sequenced to carry out the increased burden. The availability of effort will be a factor in approval. It is understood that some papers necessarily have a long maturation period because of the need to build up the data set and infrastructure. That said, JADES does not want papers that could be done promptly to be reserved yet not completed quickly due to over-commitment of the lead author.

The SC can revoke authorization for a paper if it judges that there is not a reasonable path to completion, e.g., due to lack of progress or loss of leadership of the paper.

## Expectations of Lead Authors during Analysis

Following the approval of the paper, the lead author will drive the analysis and drafting of the paper. It is important that the leads be pro-active about communicating their progress to the rest of the group; remember that many people in JADES plan to be co-authors and thereby have an expectation of transparency regarding the work. We encourage that each paper creates a clearinghouse web location to record progress, be it an Overleaf document, a Confluence page, a Google doc, or some other mechanism. We then encourage that each paper designates a Slack channel (either a new one or an existing one) for communication between the co-authors. The location of the clearing house page should be pinned on this Slack channel and included in the relevant column of the JADES list of approved papers.

Lead authors should plan to report their progress on JADES telecons, typically every 2 months. They must also respond to calls from the Publication Coordinators and/or Steering Committee for status updates.

Lead authors should be in communication with the Project Lead(s) and Publication Coordinators about substantive changes of plan, e.g., delays to technical concerns or unavailability of effort, or changes in scope of the abstract. If necessary, the Publication Coordinator will help the lead authors update the scope of the abstract and perhaps generate additional paper proposals (following the declaration process, but hopefully more quickly as more advanced work is usually understood more crisply).

Lead authors should attempt to be inclusive of other JADES members in the work. At any time during the preparation of the paper, any JADES collaboration member can still express the desire to join a given paper team (e-mail to be sent to the Lead Author), indicating how they might help with the paper. Should there be any conflict between the proposed contribution and the contribution of an existing paper team member, precedence will be given to the latter.

Lead authors are responsible to ensure that the author list is complete and that the right acknowledgements and the relevant citation to instrumental and technical papers are included. A list of reference instrumental and technical papers will be maintained by the Publication Coordinator(s) on the JADES Confluence/Jira site.

Lead Authors are responsible for conducting their work in a manner that will be consistent with data availability of the journals and funding agencies. In particular, the data in figures and tables must be readied to be provided in electronic form no later than journal publication. JADES also expects that work be conducted with reasonable attention to code versioning, data provenance, and reproducibility.

Lead Authors should discuss with the Steering Committee how the data displayed in figures and tables, and hence released upon publication, might affect other JADES papers, e.g., because the included data would support a broader utilization. This may lead to requirements from the SC to be limited in how the data are revealed.

## Collaboration-wide Review and Paper Submission

As the paper nears completion and submission, the following process should be followed:

1. If fast-track status is desired, the lead author should make this request to the SC, typically about a week before the intended start of the Collaboration-Wide Review. The request should detail the rationale for why the extra time of a standard review period presents a severe obstacle for the paper.
2. The lead author will circulate a near-final draft to the collaboration to propose the start of a Collaboration-Wide Review (CWR). The draft should be circulated by the general email list and by the #general Slack channel, including a PDF with line numbering, and an Overleaf link (if applicable). It is important that the circulated draft be well polished and complete; any deviations from a submittable version should be clearly marked. The lead author will additionally send a mail to SC ([jades-steering-committee-group@ucsc.edu](mailto:jades-steering-committee-group@ucsc.edu)) to request the start of CWR.
  - a. As part of this circulation, the lead author should float a clear proposal for the associated release of data (beyond the electronic forms of tables and figures already required by the previous section), including the description and proposed timing of the release. It is not required that this proposal be pre-approved by the SC; rather the intent is to allow a collaboration comment period and to discuss with the SC during the CWR, resulting in approval before submission.
3. The Publication Coordinators and/or designated member(s) of the SC will quickly assess the readiness of the paper to enter CWR. Unpolished drafts are inefficient for co-authors to review, and incomplete drafts are not fair to those considering co-authorship. If the paper is judged ready, then the Publication Coordinators/SC will announce the start of CWR, updating the status on the JADES list of papers, and set a deadline for comments and co-authorship requests. The deadline must be at least 2 weeks for standard papers and 1 week for fast-track papers; the SC can approve exceptions but only with very urgent cause (such as new data sets with zero proprietary period). The Publication Coordinators/SC may decide that a paper is not sufficiently complete to begin the CWR and give feedback to the lead author as to what needs to be improved.
4. It is strongly encouraged that the lead author presents the paper in a JADES telecon, preferably a pan-JADES call, either just before or during the CWR, to raise awareness of the paper and socialize the results, conclusions, and data release plans.

5. During CWR, the authors receive comments from the collaboration. Feedback from the Project Leads is particularly important and should be sought out if not forthcoming.
6. During CWR, collaboration members can request co-authorship using the co-authorship spreadsheet.
7. At the end of CWR, the authors should revise the paper in light of all comments. Detailed referee-style responses to each comment are not required, but it is reasonable to engage in some conversation regarding more complicated comments. The authors also compile the final author list, including the funding acknowledgements. The Publication Coordinators are available to advise on any areas of concern.
8. The lead author circulates the final draft to the collaboration (again, email and Slack #general) with a request to the SC that the paper be authorized for submission, indicating a proposed submission time. A period of 2 business days should be given for any final urgent comments and/or authorship concerns. During these 2 days, two members of the SC will review the paper for any programmatic concerns and will communicate authorization to the lead author to submit at the proposed time.
9. The lead author is welcome to submit to arXiv after journal submission. This timing is at the discretion of the core author group, but may need to be coordinated with any press activities.

## Release of Data

It is the responsibility of the Lead Author that the data in figures and tables be readied to be provided in electronic form no later than journal publication. Beyond this, it can happen that the authors seek to release additional data as part of a paper, e.g., details about a sample or a set of spectra, or the data set resulting from analysis of a set of objects, or data to help with reproducibility of the work.

It is important to consider the implications of the scope and timing of such release on other JADES papers in development and on the ability of other team members to pursue follow-on papers. Particularly for large scopes, there can be further intersection with the main survey data releases. The Lead Author must raise these proposals to the Steering Committee for approval, and the SC expects that Lead Author will use the mechanisms in the previous section to socialize the proposal within the broader team.

## Pre-publication Confidentiality

JADES collaboration members, including co-authors, should be mindful regarding communications with external persons that might disclose the internal discussions, pre-publication results, and/or intellectual property of the collaboration in a manner that could create competitive advantages. For example, it could be that other team members, unknown to the Lead Author and JADES leadership, may be interested in writing follow-on papers about a result and therefore have an interest in preserving the privacy until publication. The CWR period would be a good time to search for such interests. Please consult with the Lead Author, Steering Committee, and/or Publication Coordinator to consider what can or should be said to those outside the collaboration.

## Revision and Resubmission of Papers

Following receipt of a referee report, the lead author should promptly circulate the report to the full authorship list. The core authorship group takes the lead on revising the manuscript. When ready, the lead author should circulate the new version and the reply to the referee to the full authorship list and set a reasonable period for comments. This period can be short (a few business days) for simple reports but might be longer if the results/conclusions of the paper have been substantially updated. The lead author should be sensitive to the fact that all of the co-authors bear responsibility for the conclusions of the paper.

There is not a need for a formal second CWR nor SC approval for resubmission, unless the paper has been rejected and is going to another journal with major changes.

## Authorship on JADES Papers

The JADES program is the product of investment from dozens of people over many years. It is anticipated that most JADES papers will have an author list reflecting these contributions.

By default, the author list will have a two-group format. A first group of co-authors is formed by the individuals who have made a significant contribution to this specific scientific investigation (the project team). The order is decided by the Lead Author after discussion with the project team and should reflect the contribution of the various co-authors. It is typically not alphabetical. Inclusion in the first group of co-authors will require a substantial investment of time (e.g., several weeks of effort, or consistent detailed advising of a junior lead author) on that specific paper. We note that activity in a working group or a technical / scientific project or membership to a work package does not itself justify inclusion in the first set of authors, nor does providing comments on an advanced draft of the paper.

A second group is formed by the JADES team members not included in the first list who have requested co-authorship. They are listed in alphabetical order. As with any paper, authors should meet a reasonable standard of contribution to the paper, either specifically to the context of the paper or to the JADES project as a whole. The latter has been codified by the NIRCam and NIRSpec PIs through the identification of Builders; these Builders have standing to co-sign any JADES paper. The SC maintains a list of Builders, updating it as additional Builders are approved by the PIs.

For key papers, the two PIs will reserve the right to add a list of instrument team members (beyond JADES) to the author list (in the second group). This will be discussed with Lead Author.

Whenever a paper relies heavily on data processing or tools developed in the framework of JADES, the Lead Author should make sure that JADES team members who have invested efforts in the development of these tools get the appropriate credit and authorship level in the paper.

At any time prior to submission, a co-author can ask to be removed from the author list. This must be done by informing by e-mail the Lead Author and the Publication Coordinator(s) of her/his withdrawal, explaining why she/he is asking to be removed from the author list.

Page charges will be carried by the institute of the Lead Author, unless stated and agreed otherwise.

## Seminars, conference contributions and talks

Data and results originating from the JADES survey observations and work cannot be used in seminars, conferences, workshops, or any other type of public talks before they have been published or posted in a publicly accessible archive (e.g. astro-ph), unless permission is granted by the lead author of the relevant project **and** the JADES steering committee or Publication Coordinator.

We caution JADES team members that all slides presented in public talks are at risk of being captured into a web accessible form, possibly without permission of the presenter.

JADES does not expect that conference contributions will carry a long author list; this is often inappropriate because the contribution may not match closely to a published paper and will typically represent the particular views of the presenter. Of course, reference to participation in the JADES collaboration is welcome, as are citations to any journal papers that are the primary source of the results.

## Conflict resolution

Conflicts should be ideally solved through discussions with the Publication Coordinator(s). If this is not possible and a satisfactory solution for the concerned parties is not found, the issue should be raised by the Publication Coordinator(s) to the JADES steering committee, with the PIs having the last word on its resolution.

The NIRCam and NIRSpect principal investigators reserve the right to intervene directly in exceptional circumstances.



## Appendix A – Agreed distribution of leadership between the NIRCam and NIRSpec teams

Analyses often are more closely related to either imaging or spectroscopy, and in these cases the collaboration anticipates that the lead author(s) will come from that instrument team. During the founding meeting of the JADES collaboration, a straw-man distribution of leadership between the two teams per scientific topic was agreed and is listed below. Team members from the secondary team may participate in a paper but cannot be the lead author.

It is understood that this distribution only applies to the first generation of JADES papers. We expect it to evolve for the subsequent generations of papers as the scientific investigations will become less and less focused on the results of a given instrument and correspond more and more to synthesis papers. Hence, these subsequent generations of papers are open to be led by any JADES member, regardless of instrument team membership. As of this writing in August 2023, we believe that all first-generation papers have been declared as abstracts and that the team can move forward to cross-team paper leadership, while respecting the priority of the previous abstracts.

### HIGHEST REDSHIFT SOURCES – IDENTIFICATION

- Photometric redshifts – Highest redshift sources
  - Leadership: NIRCam team.
  - Includes the calibration of the photometric redshifts based on NIRSpec spectra obtained as part of the 1st epoch NIRSpec UDF deep.
- Spectroscopic confirmation of the highest redshift sources.
  - Leadership: NIRSpec team.

### GALAXY PROPERTIES

- Luminosity function, size, morphology / structure.
  - Leadership: NIRCam team.
- Merger signatures and role of mergers in the growth of galaxies.
  - Morphology.
    - Leadership: NIRCam team.
  - Kinematics of extended objects.
    - Leadership: NIRSpec team.
- Masses, ages, stellar populations (including history), stellar metallicities, population III signatures.
  - From imaging data:
    - Leadership: NIRCam team.
    - Typically going fainter and with larger statistics than the studies based on spectral data.
    - Includes UV spectral slopes.
  - From spectral characterization.
    - Leadership: NIRSpec team.
    - Includes the impact of emission-lines on these studies.

- Star formation rates.
  - Based on (UV) continuum studies.
    - Leadership: NIRCам team.
  - Based on emission-lines.
    - Leadership: NIRSpec team.
  - It is expected that at some point one or several overarching papers comparing the different methods will be prepared. They are considered a second generation papers without a pre-assigned leadership.
- Metallicities / ISM properties (line ratios).
  - Leadership: NIRSpec team.
- Outflows / winds / kinematics / dynamics.
  - Leadership: NIRSpec team.
  - Possibly morphological information on some medium NIRCам filters.

### **AGNs**

- Identification and characterization based on line ratio diagnostics.
  - Leadership: NIRSpec team.
- Matching of sources with surveys at other wavelengths (radio, X-rays).
  - Leadership: NIRCам team.
- Black-hole mass from line dispersion / BLR.
  - Leadership: NIRSpec team.
- Characterization of AGN host galaxies.
  - Based on imaging
    - Leadership: NIRCам team
  - Based on spectroscopy
    - Leadership: NIRSpec team

### **REIONISATION**

- UV photon census.
  - Leadership: NIRCам team.
- Emission-line tests (Ly continuum escape fraction).
  - Leadership: NIRSpec team.
- Ly alpha disappearance investigation through spectroscopic studies.
  - Leadership: NIRSpec team.

### **LARGE-SCALE STRUCTURE & SMALL SCALE (CLOSE PAIRS)**

- Close pair statistics.
  - From spatial distribution + photometric redshifts.
    - Leadership: NIRCам team.
  - From velocity offsets.
    - Leadership: NIRSpec team.
  - Redshift spikes.
    - Leadership: mainly the NIRSpec team.

## **MISCELLANEOUS / TRANSIENTS**

- Transients - Discovery of a transient by comparison with images from an earlier epoch.
  - Leadership: NIRCам team.
- Transient - Serendipitous discovery of a transient through spectroscopic identification.
  - Leadership: mainly the NIRSpec team but will clearly involve both teams.
- Identification of objects with extreme colors using medium versus wide filter comparison.
  - Leadership: NIRCам team.

## Appendix B – Template for paper proposal

**Title:** *paper title*

**Paper category:** *one of the paper categories listed in the publication policy document, usually “standard”.*

**Lead author:** *name of the first author of the paper*

**Paper team:** *list of the paper team members.*

**Technical or scientific project(s):** *name of the project(s) most directly associated to this paper. It is preferred to pick a single primary project, but we understand that some papers will be hybrid.*

**External collaborator flag:** *if one or several external collaborators will contribute to the paper, this should be flagged here. Membership approval by the JADES steering committee should have been obtained prior to the submission of this request. It is critical to document intellectual-property entanglements with non-JADES persons and projects at this point.*

**Abstract:** *short paper description (<400 characters).*

**Data set:** *document what data set will be utilized, e.g., data collected through May 2023.*

**Timeline:** *timeline with clear milestones for the preparation and submission of the paper.*

**Declaration of Availability of Effort:** *Describe the availability of core authors to complete the paper on the proposed time scale. This is particularly important if the lead author is committed to other JADES papers.*